

■ Real-World Data Project Retail Sales Analysis & Prediction

Thiranex Internship Project — Finance, Health or Retail
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1. Introduction

This report presents an end-to-end data analysis and sales prediction project using a real-world Retail Sales dataset from a global superstore. The project applies data science techniques including data cleaning, exploratory analysis, visualization, and machine learning to uncover business insights and predict future sales. The dataset contains 9,800 orders across multiple categories, regions, and customer segments.

2. Dataset Overview

Attribute	Details
Dataset	Global Superstore Retail Sales
Total Records	9,789 orders
Total Columns	18 columns
Date Range	2015 — 2018
Categories	Furniture, Office Supplies, Technology
Regions	East, West, Central, South
Customer Segments	Consumer, Corporate, Home Office

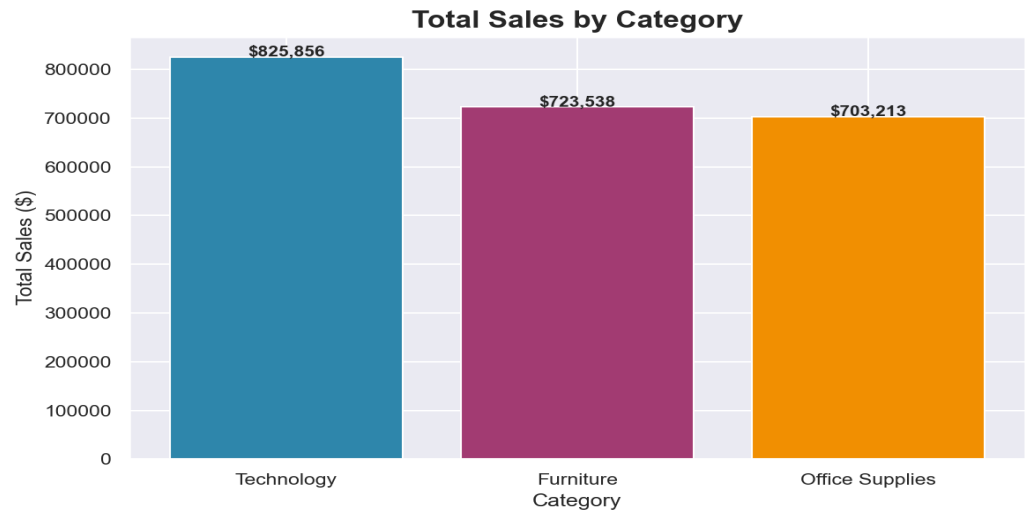
3. Statistical Summary

Metric	Value
Total Revenue	\$2,252,607.41
Average Sale Amount	\$230.12
Highest Single Sale	\$22,638.48
Lowest Single Sale	\$0.44
Total Unique Customers	793

Total Unique Products	1,860
Average Days to Ship	4.0 days
Most Popular Category	Technology
Top Region by Sales	West

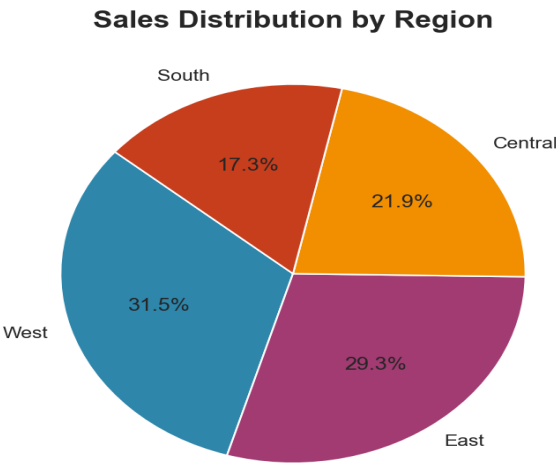
4. Visualizations & Insights

4.1 Sales by Category



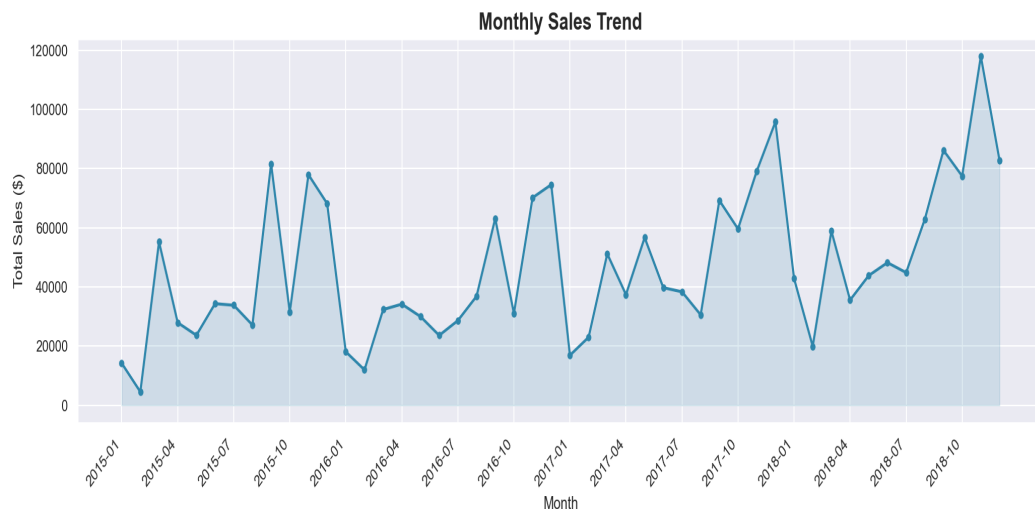
■ *Insight: Technology leads in total revenue, followed by Furniture and Office Supplies. Technology products have higher unit prices which drives the total sales value.*

4.2 Sales by Region



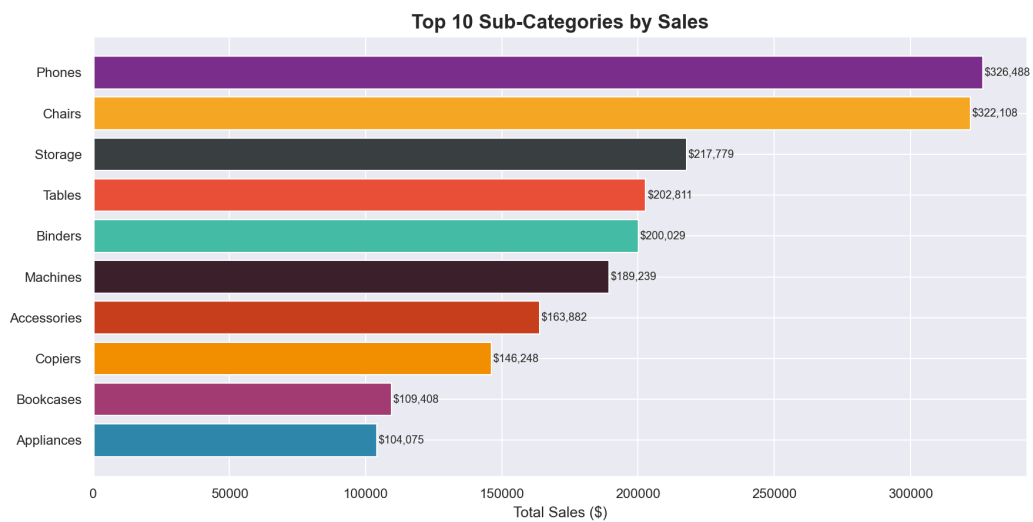
■ *Insight: The West region contributes the highest share of total sales, followed by the East. The Central region has the lowest contribution, indicating a potential growth opportunity.*

4.3 Monthly Sales Trend



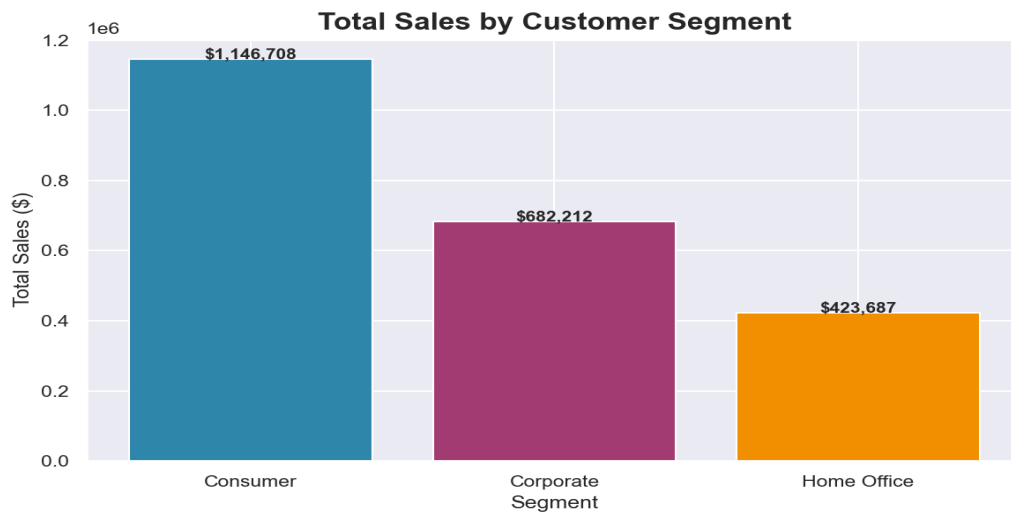
■ *Insight: Sales show a clear upward trend over the years with seasonal peaks towards the end of each year (Q4), likely driven by holiday shopping and year-end business purchases.*

4.4 Top Sub-Categories by Sales



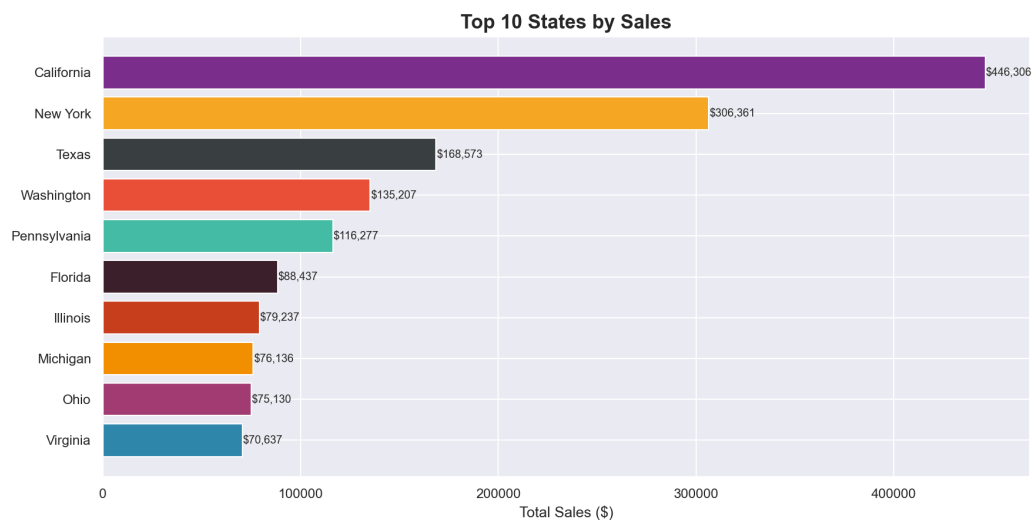
■ *Insight: Phones and Chairs are the top-selling sub-categories, reflecting strong demand for technology and office furniture in the corporate segment.*

4.5 Sales by Customer Segment



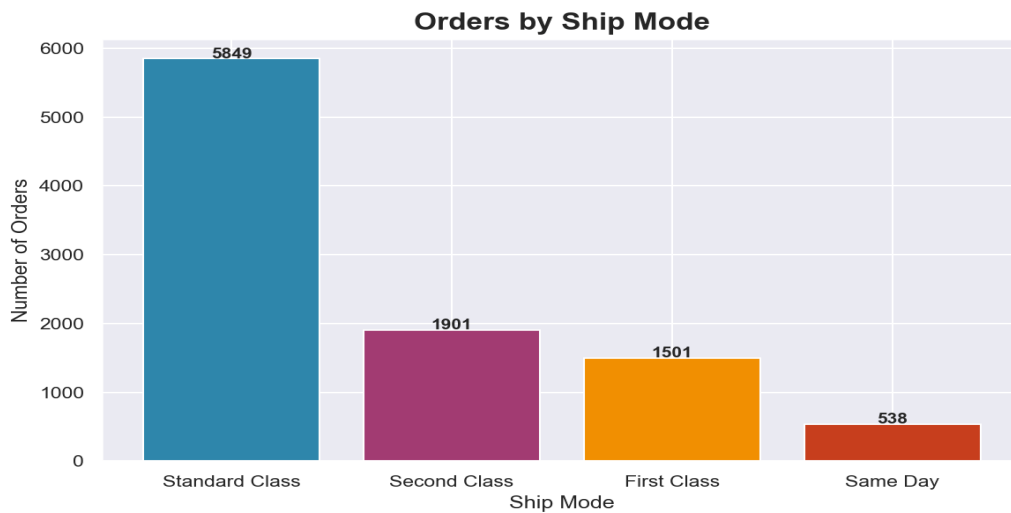
■ *Insight: The Consumer segment generates the most revenue, followed by Corporate and Home Office. Targeted marketing for Corporate clients could significantly boost revenue.*

4.6 Top 10 States by Sales



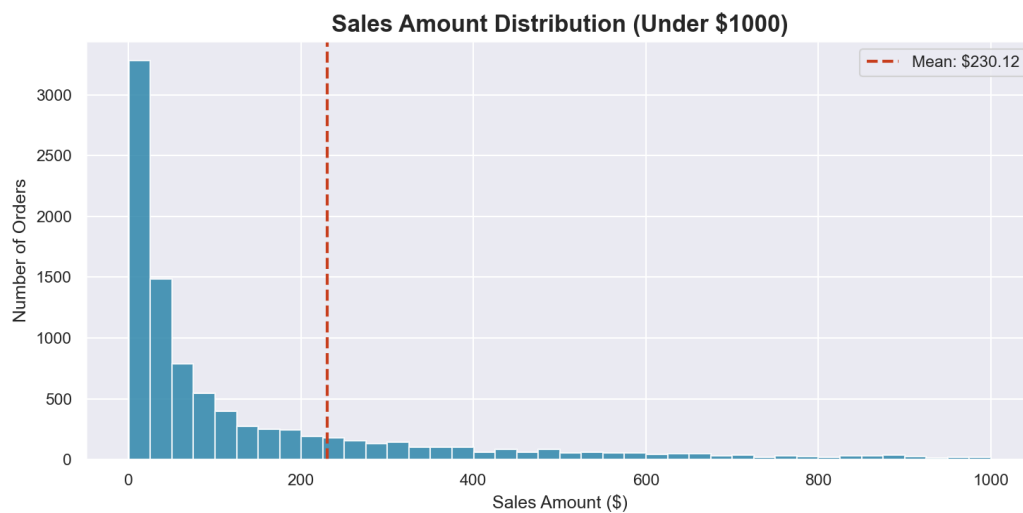
■ *Insight: California leads all states by a wide margin, followed by New York and Texas. These three states alone account for a significant portion of total national revenue.*

4.7 Orders by Ship Mode



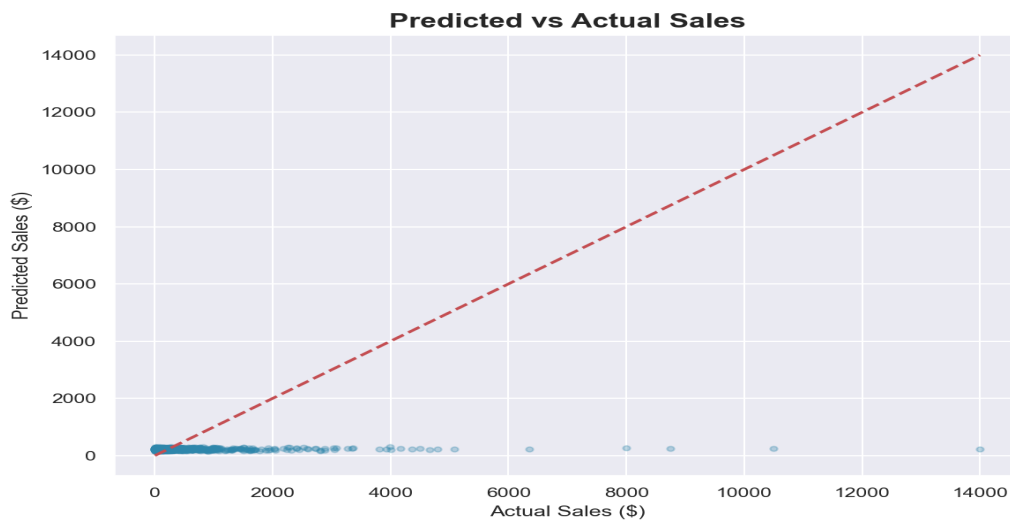
■ *Insight: Standard Class is the most preferred shipping mode, used for majority of orders. Same Day delivery is least used, suggesting customers prioritize cost over speed.*

4.8 Sales Amount Distribution



■ *Insight: Most orders fall below \$500, with the distribution heavily right-skewed. A small number of high-value orders significantly impact the total revenue figures.*

4.9 Predicted vs Actual Sales



■ *Insight: The Linear Regression model captures the general trend of sales. The spread around the diagonal line indicates room for improvement with more advanced models like Random Forest.*

5. Sales Prediction Model

A Linear Regression model was trained to predict sales amounts based on features including product category, customer segment, region, shipping mode, month, year, and days to ship. The dataset was split 80/20 for training and testing.

Metric	Value
Algorithm	Linear Regression
Training Split	80% Train / 20% Test
RMSE	\$699.62
R ² Score	-0.0007
Features Used	Category, Segment, Region, Ship Mode, Month, Year, Days to Ship

The low R² score indicates that sales are influenced by many other factors beyond the available features, such as promotions, discounts, and market trends. Future improvements could include using Random Forest or XGBoost models with additional features like discount rates and profit margins.

6. Key Findings

- Technology is the highest revenue-generating category with strong unit prices.
- The West region leads all regions in total sales contribution.
- Sales follow a seasonal pattern with peaks in Q4 each year.
- Phones and Chairs are the top-selling sub-categories by revenue.
- California, New York, and Texas are the top 3 states by sales.
- Most orders are below \$500, but high-value orders drive total revenue.
- Standard Class shipping is preferred by the majority of customers.

- Consumer segment generates the most revenue among all customer segments.

7. Conclusion

This project successfully demonstrated an end-to-end real-world data analysis on a retail sales dataset. Through data cleaning, statistical analysis, and 9 detailed visualizations, meaningful business insights were uncovered about customer behavior, regional performance, and product trends. A machine learning model was also built to predict sales values. This project fulfills all the requirements of the Thiranex Internship Real-World Data Project — applying data science skills in a real retail business context.